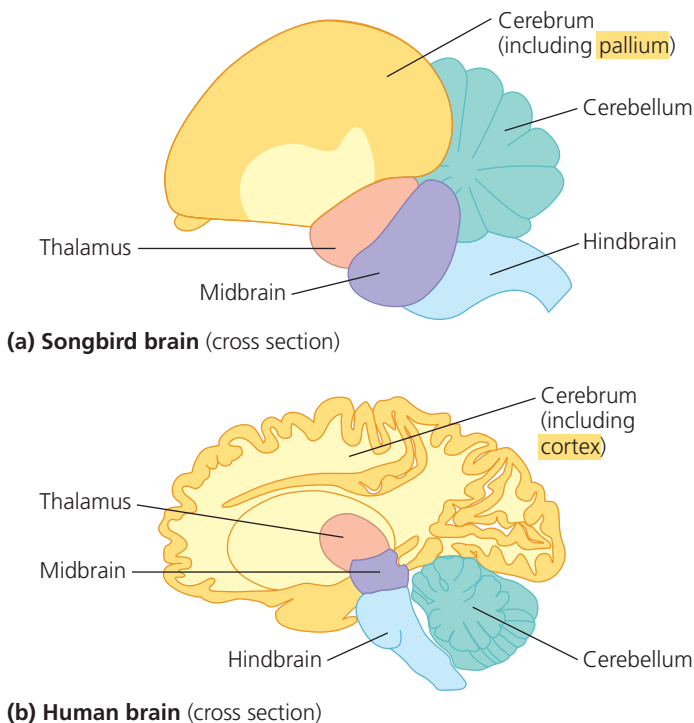


evolve in certain species? For many years researchers favored the hypothesis that higher-order reasoning in vertebrates required evolution of an extensively convoluted cerebral cortex, as is found in humans, other primates, and cetaceans (whales, dolphins, and porpoises). In humans, for example, the cerebral cortex accounts for about 80% of total brain mass.

Birds lack a convoluted cerebral cortex and were long thought to have much lower intellectual capacity than primates and cetaceans. However, recent experiments have refuted this idea: Western scrub jays (*Aphelocoma californica*) can remember which food items they hid first. New Caledonian crows (*Corvus moneduloides*) are highly skilled at making and using tools, an ability otherwise well documented only for humans and some other apes. African grey parrots (*Psittacus erithacus*) understand numerical and abstract concepts, such as “same” and “different” and “none.”

What brain structures enable some birds to have such sophisticated information processing? The answer appears to be a nuclear (clustered) organization of neurons within the *pallium*, the top or outer portion of the brain (**Figure 49.20a**). Note that this arrangement is different from that in the human cerebral cortex (**Figure 49.20b**), where six parallel layers of neurons are arranged tangential to the surface. Thus, vertebrate evolution has resulted in two different types of outer brain organization that can support complex and flexible function.

▼ **Figure 49.20 Comparison of regions for higher cognition in avian and human brains.** Although structurally different, the (a) pallium of a songbird brain and the (b) cerebral cortex of the human brain play similar roles in higher cognitive activities and make many similar connections with other brain structures.



How did the bird pallium and human cerebral cortex arise during evolution? The current consensus is that the common ancestor of birds and mammals had a pallium in which neurons were organized into nuclei, as is still found in birds. Early in mammalian evolution, this clustered organization was transformed into a layered one. However, connectivity was maintained such that, for example, the thalamus relays sensory input relating to sights, sounds, and touch to the pallium in birds and to the cerebral cortex in mammals.

Sophisticated information processing depends not only on the overall organization of a brain but also on the very small-scale changes that enable learning and encode memory. We'll turn to these changes in the context of humans in the next section.

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CONCEPT CHECK 49.3

1. How can studying individuals with damage to a particular brain region provide insight into the normal function of that region?
2. How do the functions of Broca's area and Wernicke's area each relate to the activity of the surrounding cortex?
3. **WHAT IF?** If a woman with a severed corpus callosum viewed a photograph of a familiar face, first in her left field of vision and then in her right field, why would she find it difficult to put a name to the face?

For suggested answers, see Appendix A.

CONCEPT 49.4

Changes in synaptic connections underlie memory and learning

Formation of the nervous system occurs stepwise. First, regulated gene expression and signal transduction determine where neurons form in the developing embryo. Next, neurons compete for survival. Every neuron requires growth-supporting factors, which are produced in limited quantities by tissues that direct neuron growth. Neurons that don't reach the proper locations fail to receive such factors and undergo programmed cell death. The net effect is the preferential survival of neurons that are in a proper location. The competition is so severe that half of the neurons formed in the embryo are eliminated.

In the final phase of organizing the nervous system, synapse elimination takes place. During development, each neuron forms numerous synapses, more than are required for its proper function. Once a neuron begins to function, its activity stabilizes some synapses and destabilizes others. By the time the embryo completes development, more than half of all synapses have been eliminated. In humans, this elimination of unnecessary connections, a process called synaptic pruning, continues after birth and throughout childhood.